

Being Together Without Interaction: Ambient Connection and Collectivity in a Multiplayer XR Rave Experience

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Figure 1: In Pursuit of Repetitive Beats.

ABSTRACT

This paper examines how extended reality (XR) can enable social presence and collective engagement without explicit interaction. Using *In Pursuit of Repetitive Beats* (IPORB), a multi-sensory VR experience recreating Coventry's 1989 rave scene as a case study, we combine audience evaluation data ($n = 586$) with design analysis to explore inclusion, accessibility, and emotional dynamics. IPORB achieved high scores for immersion, cultural value, and emotional intensity, suggesting that co-presence and synchronised narrative experiences may contribute to meaningful social connection even in the absence of verbal or task-based collaboration. Inclusive design features, such as low-barrier interaction, adaptable participation modes, and culturally situated representation, aim to broadened access to diverse audiences, including those new to XR. Findings highlight the role of shared emotional experience and embodied awareness in shaping socially cohesive XR environments and call for integrating broad

evaluative frameworks with targeted social XR measures in future research.

Index terms: Co-presence, Shared immersion, Audience evaluation.

1 INTRODUCTION

Extended reality (XR) technologies are increasingly being used to create shared cultural and social experiences (Roussou & Katifori, 2018) that bring audiences together within immersive environments. As these technologies become more prevalent for the public, questions around social interaction and inclusion are becoming central to how such experiences are designed (Dudley et al., 2023). While much XR research focuses on technical performance or interaction mechanics (Radianti et al., 2020), there remains a need for applied case studies that examine how social presence and emotional engagement are experienced by audiences in real-world settings.

This paper draws on an evaluation of *In Pursuit of Repetitive Beats* (IPORB) produced by East City Films (UK), a multi-sensory VR experience exploring Coventry's rave scene in 1989. The experience toured seven venues across the UK from July 2024 to October 2025, with i2 media research acting as the experience's evaluation partner. A total of 586 audience members completed the post-experience survey, enabling analysis across multiple regional contexts.

Audience responses were measured using i2 media research's proprietary Audience Impact Metric (AIM), which captures key dimensions of audience experience, including global quality, cultural value, engagement, and emotional intensity. The AIM enables experiences to be benchmarked against a wider set of immersive and cultural experiences, situating IPORB within a broader evaluative landscape rather than treating it as an isolated case.

By combining large-scale audience evaluation data with analysis of the experience's design, this paper presents IPORB as a case through which broader questions about social presence and shared emotional experience in XR can be explored. The following analysis reflects on how representation and affective response may intersect with social interaction, offering context that may inform discussions on inclusive XR design.

2 IN PURSUIT OF REPETITIVE BEATS: A CASE STUDY

According to the experience's writer and Director Darren Emerson, IPORB is distinguished by its use of VR in combination with sound design, art direction, lighting, and music to construct a historically situated and immersive audience experience. Conceived as an embodied form of music documentary, the experience places audiences within a representation of the West Midlands' illegal rave scene of the late 1980s, with the aim of evoking aspects of youth culture, musical discovery, and collective participation. The design of IPORB attempts to mirror the collective ethos of rave culture, combining shared presence and collective immersion rather than prioritising individual interaction.

The focus on immersion and historical transportation supports the experience's broader objectives of fostering social connection through engagement with shared cultural history. By exploring personal accounts and local perspectives, the experience frames the rave scene as a socially meaningful movement rooted in community formation and youth-led cultural practices, while acknowledging the complexities of its illegal nature.

Social connection is embedded in the experience not only conceptually, but structurally through its shared, multi-user design. Up to four audience members can participate simultaneously, perceiving one another as co-present participants. The experience adopts a hybrid configuration that combines physical installation elements with virtual environments developed using animation, 360-degree video, and volumetric capture. Together, these design choices aim to support an embodied and collective mode of engagement that extends beyond more traditional, individualised XR experiences.

Because of its emphasis on co-presence and shared engagement, IPORB serves as a strong case study for examining social interaction in extended reality. Unlike many XR experiences that rely on individual, disconnected head-mounted displays, IPORB is situated within a room-scale environment where participants experience content together rather than in isolation.

This format positions IPORB at the intersection of XR and live cultural experience, making it a useful context for investigating social presence and collective engagement. The experience is designed to immerse audiences in the audiovisual environment while allowing awareness of other participants in the shared physical space, supporting subtle forms of co-located interaction such as shared attention and synchronised responses. These dynamics align with research suggesting that shared immersive environments can foster heightened social presence even in the absence of verbal or gestural interaction (Son, Tiemann & Rubo, 2025).

3 EVALUATION OF IN PURSUIT OF REPETITIVE BEATS USING THE AUDIENCE IMPACT METRIC

Across all four AIM dimensions, the experience scored at or near the upper range of the benchmark database, with particularly strong performance on measures of cultural poignance, immersion, enjoyment, and positive emotional response (see Figures 1. & 2. in the Appendix).

While the AIM does not directly measure constructs such as social presence, collaboration, or inclusion, it captures affective and experiential dimensions that are often underrepresented in more technically focused XR evaluations (Lin et al., 2025). The consistency of responses across multiple venues further indicates that these outcomes were not confined to a single physical or cultural context. The following sections examine specific design features of the experience that may help explain these evaluative outcomes in relation to themes central to the workshop, including inclusion, accessibility, social presence, emotion, and cultural representation.

4 DESIGN FOR INCLUSION AND ACCESSIBILITY

In Pursuit of Repetitive Beats was designed to be accessible to audiences regardless of previous experience with XR technologies. Inclusion is supported from the outset through staff-led onboarding, during which participants receive an overview of the experience, assistance with equipment, and support throughout the session. This approach is designed to reduce uncertainty and lowers barriers for first-time users.

Interaction within the experience is intentionally limited to a small set of simple actions, such as pushing, picking up, or dropping objects using ergonomic controllers that require only trigger input. These interactions are introduced during onboarding and are visibly reinforced through minimal hand-based avatars visible to participants during the experience. The decision to minimise complex control schemes and high dexterity inputs reduces the need for prior familiarity with XR and video gaming interfaces, supporting broad accessibility. Where interactions are not completed, the experience automatically progresses, preventing participants from becoming stuck.

Physical accessibility is addressed through multiple participation modes. While IPORB is primarily a multiple room-scale experience, a seated version is available on request, enabling users to navigate the environment via virtual teleportation rather than physical movement. Seated users share the same virtual space as free-roaming users, maintaining co-presence and social interaction. Wheelchair users are similarly accommodated within the shared physical and virtual space with spatial indicators beneath avatars used to support safe and inclusive movement.

Additional accessibility measures include multilingual subtitles (English, French, Spanish, and Mandarin) and the provision of scene description packs for audiences with visual impairments, sensory concerns, and cognitive needs.

i2 media research's evaluation data indicates patterns consistent with reduced barriers to participation. 39% of survey respondents reported being new to XR, with a further 30% indicating infrequent or lacking prior exposure (see Figure 3. in the Appendix). The audience profile also differed from typical XR participation patterns, with a substantial proportion of participants aged 45 and older and a broadly balanced gender distribution leaning slightly towards women (see Figures 4. & 5. in the Appendix). This contrasts with existing research indicating that traditional XR audiences and users skew younger (Deloitte, 2023; Academy of Animated Art, 2024) and towards men (Academy of Animated Art, 2024). While multiple factors may contribute to this pattern, these findings indicate that IPORB reached beyond established XR audiences.

In summary, these features reflect an inclusive XR design approach that prioritises adaptability, user autonomy, and reduced cognitive and physical demands. This aligns with research emphasising flexible interaction modes and accessibility-oriented design as key to expanding participation in immersive experiences (Dudley et al., 2023).

5 SOCIAL PRESENCE AND EMOTION

Although IPORB does not include explicit multiplayer interaction mechanics, it supports meaningful social interaction through shared immersion and collective attention. Rather than relying on task-based collaboration, the experience facilitates a form of passive or implicit social engagement, in which audience members encounter and interpret the experience together within a shared virtual space.

Previous research indicates that synchronized sensory experiences can promote social bonding and group cohesion even when interaction remains implicit rather than explicit (Wei et al., 2024). Although social presence was not measured directly in this evaluation, the consistently high engagement and emotional intensity scores across venues may be partially related to the shared mode of participation afforded by the experience. IPORB's guided, collective temporal progression may therefore contribute to a shared emotional trajectory among audiences, reinforcing the social dimension of immersive XR.

Audience responses also reported high positivity of emotions experienced and comparatively low negativity of emotions. While the evaluation does not capture interpersonal emotional dynamics directly, prior research suggests that co-present groups may experience forms of emotional alignment through subtle behavioural cues, such as shared stillness or patterned movements (Linares-Vargas & Cieza-Mostacero, 2024). The shared physical setting of IPORB is therefore consistent with conditions that may facilitate such processes, although this cannot be confirmed from the present data.

Social presence is also supported through embodiment within the virtual environment, with avatar design playing a key role in how participants perceive one another and experience co-presence (Slater & Sanchez-Vives, 2016). Avatar design in IPORB prioritises recognisable human forms while avoiding detailed realism. Audience members are represented through abstract, ethereal avatars with tracked upper bodies and hands, enabling awareness of others' movements, without reinforcing normative, binary, or standardised gender and body forms. Colour coding and grounded positional markers further support orientation and co-presence within both the virtual and physical space.

To summarise, these elements illustrate how social presence and shared emotional experience can be supported through co-presence, embodiment, and narrative synchronisation even in the absence of explicit interaction mechanics.

The following section considers how cultural representation within the experience shapes audience identification and validation.

6 CULTURAL REPRESENTATION AND VALIDATION

Cultural representation within IPORB is addressed through the inclusion of diverse voices and perspectives, reflecting the social reality of Coventry and the West Midlands rave scene. Rather than focusing on a single socio-demographic narrative, the experience presents multiple lived experiences, reflecting the heterogeneity of the communities that shaped the scene.

Contributors featured within the documentary content include individuals from varied racial identities, gender backgrounds, and socio-professional roles, with on-screen representations similarly reflecting this diversity. By presenting culturally recognisable individuals rather than abstract or generic avatars, the experience provides representational cues that may facilitate identification and a sense of cultural relevance for some audience members. This approach positions the narrative as socially situated rather than purely abstract, potentially supporting forms of empathy and recognition without assuming these outcomes for all participants.

This approach aligns with broader discussions in XR research emphasising the importance of cultural sensitivity and representation in immersive environments (Dhiman, 2023; Anwar et al., 2025). In this respect, IPORB illustrates how immersive storytelling and representational choices may create conditions that support inclusive social engagement and share meaning within XR environments.

7 LIMITATIONS

This study has several limitations that should be considered when interpreting the findings. Firstly, this paper draws on AIM measures that are intentionally designed for broad applicability across diverse XR experiences. While these measures capture meaningful and relevant dimensions of experience, their design limits their sensitivity to constructs specific to social interaction and collaboration in XR contexts. In addition, the evaluation employs proprietary, non-standardised scales rather than established instruments used to measure constructs such as social presence, co-presence, and collaborative engagement. Although these measures are benchmarked against other XR experiences, their closed and non-public nature limits external transparency, comparability and independent replication. Finally, this work is presented as a case study of a specific XR experience and its design approach. Despite drawing from a large sample and multiple venues, the findings may

not be generalisable to all immersive XR contexts, particularly those that are explicitly interactive. Despite these limitations, the study offers context for considering how shared immersion and inclusive design approaches may relate to social connection in XR experiences.

8 CONCLUSION

This paper has examined *In Pursuit of Repetitive Beats* as a case study for considering social interaction and emotional engagement within an immersive XR experience. By drawing on audience evaluation data alongside analysis of the experience's design, the paper explores how shared immersion and co-presence may relate to social presence, even in the absence of explicit verbal or task-based interaction. Within IPORB, collective attention, embodied awareness, and a synchronised narrative structure appear to offer conditions under which audience members may experience forms of social engagement.

The experience also illustrates how inclusive design choices, including low-barrier interaction, adaptable physical participation and culturally situated representation, may contribute to broadening the accessibility and relevance of public-facing-XR. Patterns observed in the evaluation, including strong affective responses and engagement from audiences with varied levels of XR familiarity, are consistent with the possibility that shared emotional experience plays a role in shaping socially cohesive immersive environments, although this cannot be confirmed from the present data.

While social presence and co-presence are well-established areas of XR research, the evaluation presented here was not designed to isolate or measure these constructs directly. Instead, it employed a broad, experience-focused framework intended to capture overall audience response. As such, the findings highlight the value of integrating large-scale, descriptive evaluative approaches with more targeted social-XR measures in future work, particularly where shared immersion and co-presence are central design features. Such integration would support a more precise understanding of social and emotional dynamics within public-facing, collaborative XR experiences.

SUPPLEMENTAL MATERIALS

Supplemental materials including the appendix are provided as part of the submission.

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